

MEASURE: Multi-Task Slice Selection and Regression for Neonatal Brain Biometry

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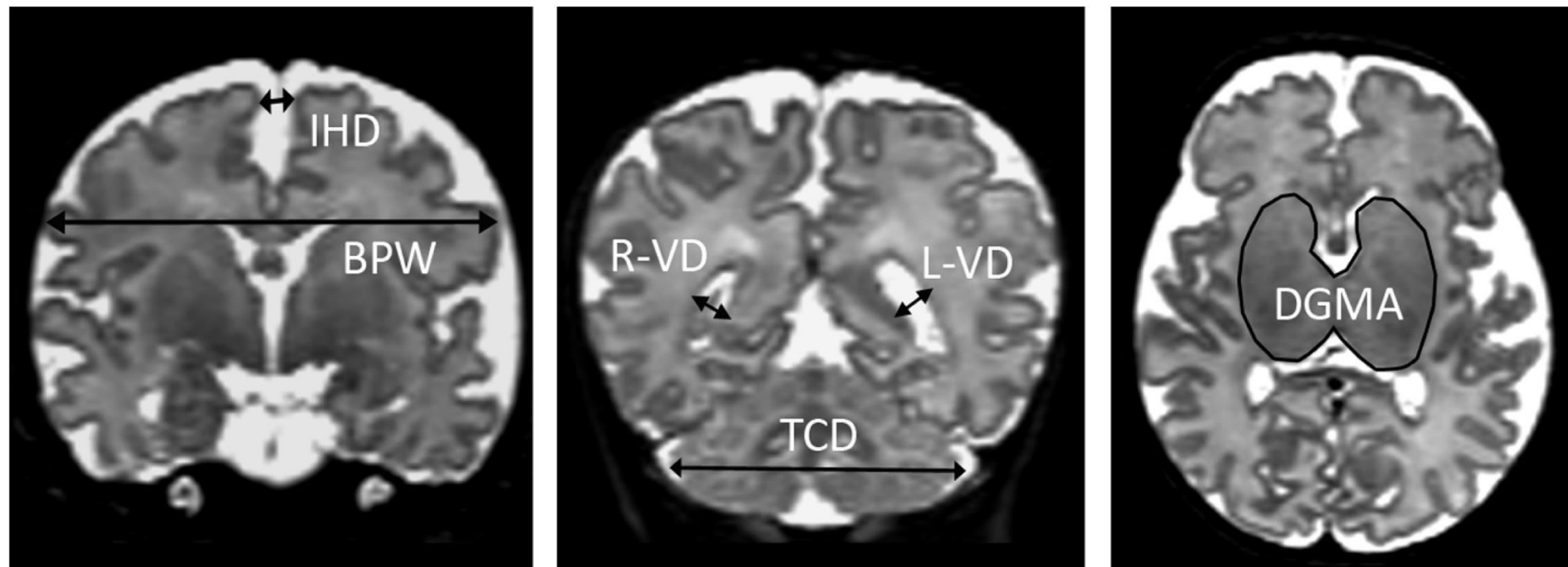


TL;DR: Six standard brain-size measurements from newborn MRI : find the right slice, then measure, no pixel-level labels needed.

MOTIVATIONS

Background

- Kidokoro biometry on term-equivalent MRI provides objective markers of brain growth and injury in preterm infants.
- Manual measurement takes 15–30 min/scan and varies between raters.
- Existing automated methods **often require dense segmentation or landmark labels, increasing annotation burden and fragility under domain shift.**



H Kidokoro et al., Kidokoro quantitative brain biometry assessments

Research Purpose

- Develop MEASURE: a clinically aligned system that predicts six Kidokoro biometrics from dual-view T2 MRI.
- Core idea: **select measurement planes first, then jointly regress biometrics without dense annotations.**

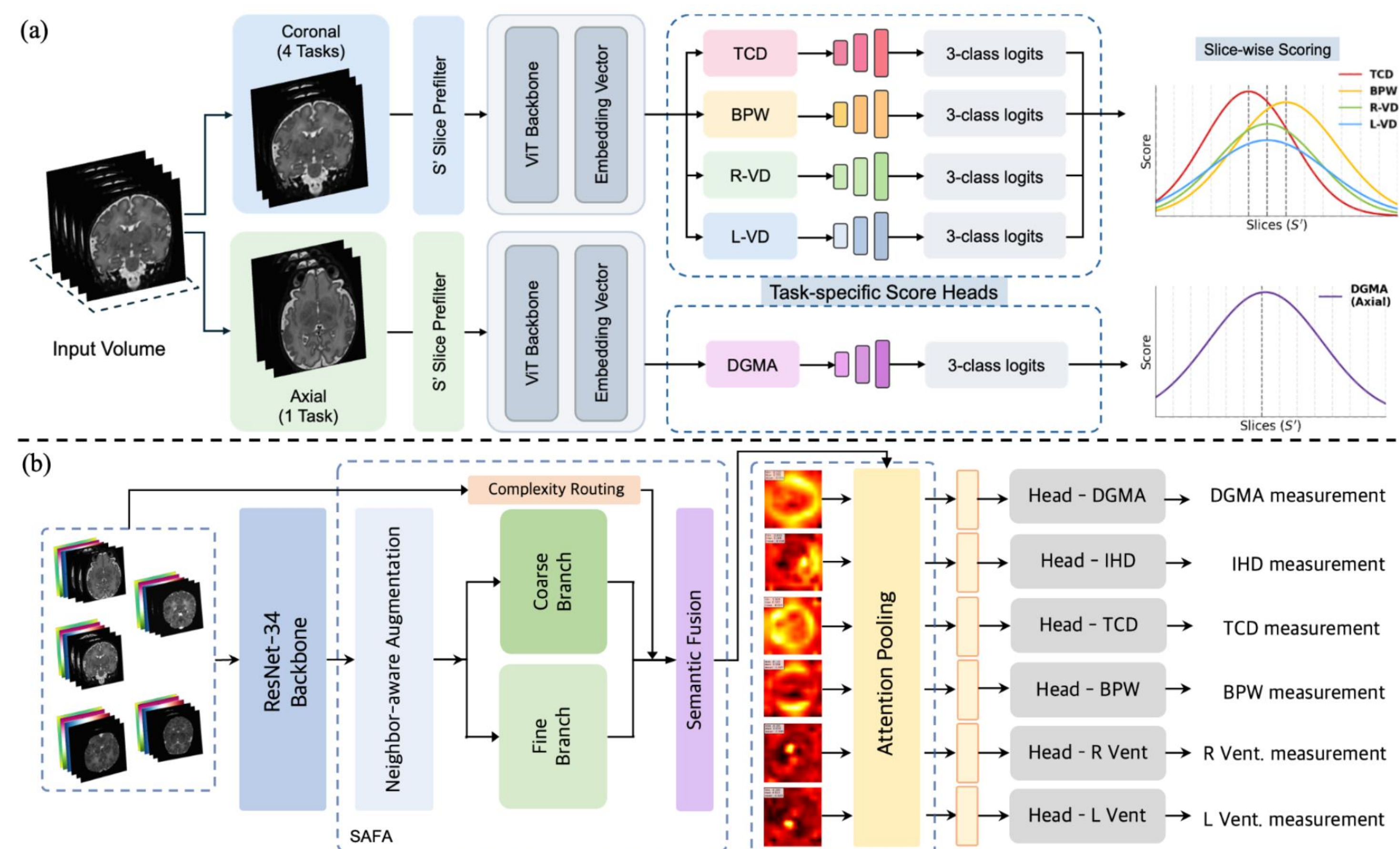
METHOD

Stage A : Slice Selector

- 2.5D input: adjacent slices (s-1, s, s+1) from axial and coronal views
- ViT scores every slice using a task-specific 3-class head (strong/weak/negative)
- Outputs the Top-K informative slices for each biometric

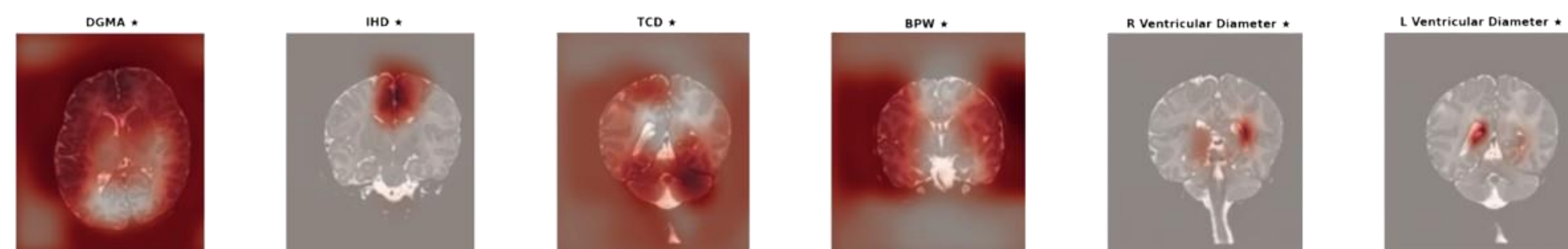
Stage B : Predictor

- A ResNet-34 encodes the selected slice and its neighbors with 3 normalized coordinate channels
- Inter-slice differences capture through-plane anatomical context
- Fine- and coarse-scale branches are adaptively fused pixel-wise
- A Sobel-driven routing mask emphasizes regions according to local edge complexity



Learnable Spatial Importance Pooling

- SIX biometric-specific attention maps
- Anatomy-aware spatial pooling, **regularized for ventricular symmetry, smoothness, and inter-task distinctness**



Able to learn the correct anatomy without voxel-level supervision

RESULTS & CONCLUSION

Baselines

- No deep-learning method exists for neonatal biometry, so we compare against an age-only lower bound, 3D models (to test whether explicit plane selection helps), multi-task regressors from related clinical tasks.
- **PMA regression:** predicts biometrics from age alone, the **lower bound.**
- **3D volumetric models:** SwinUNETR-V2, 3D ViT and BrainAge regress end-to-end from the whole volume, with **no slice selection.**
- **2D multi-task models:** MECDS and SePL adapted from other clinical tasks and use the slices chosen by our selector.

Dataset

- 470 **term-equivalent neonates** (36–42 weeks PMA) from dHCP with 3T T2-weighted MRI.
- Scans degraded to **simulate clinical 5 mm thick slices.**
- Six biometrics measured **manually by an expert clinician**, evaluated with **5-fold subject-level cross-validation.**

Results

Method	Type	ICC $\uparrow$						Pearson $r \uparrow$						MAE $\downarrow$					
		TCD	BPW	R-VD	L-VD	DGMA	IHD	TCD	BPW	R-VD	L-VD	DGMA	IHD	TCD	BPW	R-VD	L-VD	DGMA	IHD
Comparison Methods																			
PMA	–	.697	.722	.052	.222	.367	.032	.730	.735	.060	.334	.463	.149	1.80	4.88	1.10	0.91	0.85	0.86
SUNETRV2	3D	.649	.787	.104	.094	.758	.101	.690	.807	.192	.172	.781	.229	2.00	2.68	0.93	1.08	0.55	0.80
BrainAge	3D	.780	.903	.459	.552	.821	.416	.805	.909	.505	.588	.835	.445	1.64	1.84	0.82	0.92	0.49	0.83
ViT3D	3D	.675	.812	.175	.259	.731	.161	.710	.825	.246	.358	.750	.272	1.97	2.63	0.94	1.02	0.59	0.80
MECDS [‡]	2D	.816	.907	.622	.673	.814	.418	.853	.915	.666	.712	.832	.479	1.43	1.88	0.69	0.77	0.48	0.77
SePL [‡]	2D	.800	.909	.629	.660	.824	.352	.842	.924	.673	.719	.848	.438	1.50	1.76	0.69	0.78	0.47	0.76
MEASURE (Ours)	2D	.847*	.924*	.674*	.722*	.837	.485*	.880*	.930	.711*	.759*	.855	.564*	1.33*	1.62*	0.66	0.73	0.46	0.71†
Ablation Study (MEASURE variants)																			
Base	–	.734	.904	.447	.558	.806	.367	.763	.912	.511	.627	.821	.424	1.73	1.88	0.82	0.88	0.51	0.81
w/o Attn Pool & AAR	–	.828	.919	.596	.673	.833	.502	.840	.920	.635	.702	.841	.525	1.52	1.81	0.73	0.80	0.48	0.76
w/o SAFA	–	.857°	.920	.699°	.748°	.829	.459	.870	.924	.718°	.770°	.836	.500	1.34	1.77	0.64°	0.71°	0.48	0.76
Full Model	–	.847	.924	.674	.722	.837	.486	.880	.930	.711	.759	.855	.564	1.33	1.62†	0.66	0.73	0.46	0.71†

- Best on biometrics across ICC, pearson r and MAE; significant for most.
- Explicit slice selection-based methods beat 3D volumetric ones.
- Spatial pooling is the component that helps most.

Measure	Human BLoA	Model BLoA	LoA Ratio	Coverage
BPW	−0.24 (−7.77; 7.30)	0.71 (−4.37; 5.79)	0.66 (0.45; 1.10)	97.7%
IHD	−0.32 (−2.22; 1.57)	−0.01 (−1.84; 1.82)	0.99 (0.77; 1.29)	90.8%
TCD	0.17 (−2.47; 2.80)	0.12 (−3.29; 3.53)	1.28 (1.04; 1.64)	88.5%
DGMA	−0.13 (−1.39; 1.13)	−0.11 (−1.36; 1.14)	1.17 (0.90; 1.48)	90.8%
L-VD	−0.37 (−3.04; 2.30)	0.05 (−1.92; 2.03)	0.73 (0.62; 0.87)	97.7%
R-VD	−0.30 (−2.79; 2.18)	0.09 (−1.60; 1.69)	0.68 (0.54; 0.88)	98.9%

- Bland-Altman analysis: model vs. human intra-rater variability (3-month interval)
- LoA Ratio < 1 indicates model is more consistent than human re-measurement.

CROSS DOMAIN EVALUATION

FeTA 2024

- We transferred MEASURE to fetal biometry (FeTA 2024, 70 subjects), keeping the core architecture and replacing only the prediction heads.
- Adding the image to GA improved the large structures, while image alone could not beat GA, consistent with the FeTA 2024 challenge.

Model	MAE (mm) ↓					ICC ↑				
	bBIP	sBIP	HV	LCC	TCD	bBIP	sBIP	HV	LCC	TCD
GA only	2.91	3.50	1.27	2.84	2.55	0.937	0.916	0.817	0.809	0.895
Image only	3.69	4.34	1.52	2.98	2.52	0.852	0.835	0.751	0.749	0.862
Image + GA	2.88	3.16	1.51	2.91	2.09	0.932	0.921	0.729	0.799	0.916

Cross-domain evaluation on the FeTA 2024 dataset using MAE and ICC



http://cna.hanyang.ac.kr/

Brain-Specialized Self-Supervised Learning: Neuroanatomy-Aware Spatial Priors for 3D Brain MRI

Kyeong Ho Kim, Minkyong Kwon et al., 2026 MICCAI

Poster Session 3, Board Number T-PM-170

Another work from our Lab.